

ADM-HQ:

- Находясь с требуемой директории с активированным виртуальным окружением
 - создаём файл **playbook1_keepalived.yml**
 - и описываем весь необходимый функционал

```
cat << EOF > playbook1_keepalived.yml
- name: Install and settings keepalived for HA1-COD and HA2-COD
  hosts: proxy
  become: true

  tasks:
    - name: Install package 'keepalived'
      community.general.apt_rpm:
        name: "keepalived"
        state: present
        update_cache: true

- hosts: ha1-cod
  become: true

  tasks:
    - name: Copy the 'keepalived.conf' file for MASTER
      ansible.builtin.template:
        src: templates/keepalived-master.conf.j2
        dest: /etc/keepalived/keepalived.conf
        owner: root
        group: root
        mode: '0644'

- hosts: ha2-cod
  become: true

  tasks:
    - name: Copy the 'keepalived.conf' file for BACKUP
      ansible.builtin.template:
        src: templates/keepalived-backup.conf.j2
        dest: /etc/keepalived/keepalived.conf
        owner: root
        group: root
        mode: '0644'

- hosts: proxy
  become: true

  tasks:
    - name: Started and enabled keepalived
      ansible.builtin.systemd:
        name: keepalived
        state: started
        enabled: true

EOF
```

- Создаём директорию для шаблонов:

```
mkdir templates
```

- Создадим сами шаблоны:

```
cat <<EOF > templates/keepalived-master.conf.j2
global_defs {
    enable_script_security
    max_auto_priority
}

vrrp_script chk_haproxy {
    script "killall -0 haproxy"
    interval 2
    weight 2
}

vrrp_instance VI_1 {
    interface {{ keepalived_interface_name }}
    state MASTER

    virtual_router_id 51
    priority 101

    virtual_ipaddress {
        {{ keepalived_virtual_ipaddress }}
    }

    track_script {
        chk_haproxy
    }
}
EOF
```

```
cat <<EOF > templates/keepalived-backup.conf.j2
global_defs {
    enable_script_security
    max_auto_priority
}

vrrp_script chk_haproxy {
    script "killall -0 haproxy"
    interval 2
    weight 2
}

vrrp_instance VI_1 {
    interface {{ keepalived_interface_name }}
    state BACKUP

    virtual_router_id 51
    priority 100

    virtual_ipaddress {
        {{ keepalived_virtual_ipaddress }}
    }

    track_script {
        chk_haproxy
    }
}
EOF
```

- Создадим необходимые переменные:

```
cat <<EOF > inventories/production/group_vars/proxy.yml
keepalived_interface_name: "ens19"
keepalived_virtual_ipaddress: "172.16.1.253/23"
EOF
```

- Запускаем данный playbook:

```
ansible-playbook -i inventories/production/hosts playbook1_keepalived.yml
```

- Результат:

```

(ansible) [user@adm-hq ansible]$ ansible-playbook -i inventories/production/hosts playbook1_keepalived.yml

PLAY [Install and settings keepalived for HA1-COD and HA2-COD] *****

TASK [Gathering Facts] *****
ok: [ha2-cod]
ok: [ha1-cod]

TASK [Install package 'keepalived'] *****
changed: [ha1-cod]
changed: [ha2-cod]

PLAY [ha1-cod] *****

TASK [Gathering Facts] *****
ok: [ha1-cod]

TASK [Copy the 'keepalived.conf' file for MASTER] *****
changed: [ha1-cod]

PLAY [ha2-cod] *****

TASK [Gathering Facts] *****
ok: [ha2-cod]

TASK [Copy the 'keepalived.conf' file for BACKUP] *****
changed: [ha2-cod]

PLAY [proxy] *****

TASK [Gathering Facts] *****
ok: [ha2-cod]
ok: [ha1-cod]

TASK [Started and enabled keepalived] *****
changed: [ha2-cod]
changed: [ha1-cod]

PLAY RECAP *****
ha1-cod      : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ha2-cod      : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

(ansible) [user@adm-hq ansible]$

```

- Проверить, наличие VIP на HA1-COD:

```

(ansible) [user@adm-hq ansible]$ ssh root@ha1-cod
Last login: Thu Mar 19 10:53:21 2026 from 10.0.2.1
[root@ha1-cod ~]# ip -c -br -4 a
lo                UNKNOWN    127.0.0.1/8
ens19             UP        172.16.0.1/23 172.16.1.253/23
[root@ha1-cod ~]# exit
Выход
Connection to ha1-cod closed.
(ansible) [user@adm-hq ansible]$

```

- Запускаем данный playbook повторно:
 - все задачи должны быть **OK**

```

(ansible) [user@adm-hq ansible]$ ansible-playbook -i inventories/production/hosts playbook1_keepalived.yml

PLAY [Install and settings keepalived for HA1-COD and HA2-COD] *****

TASK [Gathering Facts] *****
ok: [ha1-cod]
ok: [ha2-cod]

TASK [Install package 'keepalived'] *****
ok: [ha1-cod]
ok: [ha2-cod]

PLAY [ha1-cod] *****

TASK [Gathering Facts] *****
ok: [ha1-cod]

TASK [Copy the 'keepalived.conf' file for MASTER] *****
ok: [ha1-cod]

PLAY [ha2-cod] *****

TASK [Gathering Facts] *****
ok: [ha2-cod]

TASK [Copy the 'keepalived.conf' file for BACKUP] *****
ok: [ha2-cod]

PLAY [proxy] *****

TASK [Gathering Facts] *****
ok: [ha1-cod]
ok: [ha2-cod]

TASK [Started and enabled keepalived] *****
ok: [ha2-cod]
ok: [ha1-cod]

PLAY RECAP *****
ha1-cod      : ok=6    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ha2-cod      : ok=6    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

- Находясь с требуемой директории с активированным виртуальным окружением
 - создаём файл **playbook2_web.yml**
 - и описываем весь необходимый функционал

```
cat <<EOF > playbook2_web.yml
---
- name: Install Installing the Angie Web Server
  hosts: server
  become: true

  tasks:
    - name: Install package 'angie'
      community.general.apt_rpm:
        name: "angie"
        state: present
        update_cache: true

    - name: Copy the 'index.html' file
      ansible.builtin.template:
        src: templates/index.html.j2
        dest: /usr/share/angie/html/index.html
        owner: root
        group: root
        mode: '0644'

    - name: Started and enabled angie
      ansible.builtin.systemd:
        name: angie
        state: started
        enabled: true
EOF
```

- создаём необходимый шаблон:

```
cat <<EOF > templates/index.html.j2
<html>
  <head>
    <title>AU_Team</title>
  </head>
  <body>
    <h1>{{ ansible_facts['hostname'] }} by Angie!</h1>
  </body>
</html>
EOF
```

- Запускаем данный playbook:

```
ansible-playbook -i inventories/production/hosts playbook2_web.yml
```

- Результат:

```
(ansible) [user@adm-hq ansible]$ ansible-playbook -i inventories/production/hosts playbook2_web.yml

PLAY [Install Installing the Angie Web Server] *****

TASK [Gathering Facts] *****
ok: [srv1-cod]
ok: [srv3-cod]
ok: [srv2-cod]

TASK [Install package 'angie'] *****
changed: [srv3-cod]
changed: [srv1-cod]
changed: [srv2-cod]

TASK [Copy the 'index.html' file] *****
changed: [srv1-cod]
changed: [srv3-cod]
changed: [srv2-cod]

TASK [Started and enabled angie] *****
changed: [srv1-cod]
changed: [srv3-cod]
changed: [srv2-cod]

PLAY RECAP *****
srv1-cod      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
srv2-cod      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
srv3-cod      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

(ansible) [user@adm-hq ansible]$
```

- Проверяем доступность веб-серверов:

```
(ansible) [user@adm-hq ansible]$ curl -I http://srv1-cod.au.team
HTTP/1.1 200 OK
Server: Angie/1.10.2
Date: Thu, 19 Mar 2026 08:00:07 GMT
Content-Type: text/html
Content-Length: 120
Last-Modified: Thu, 19 Mar 2026 07:59:15 GMT
ETag: "69bbacd3-78"
Accept-Ranges: bytes
Cache-Control: Idco-NGFW;detail=mismatch
Connection: keep-alive

(ansible) [user@adm-hq ansible]$ curl -I http://srv2-cod.au.team
HTTP/1.1 200 OK
Server: Angie/1.10.2
Date: Thu, 19 Mar 2026 08:00:13 GMT
Content-Type: text/html
Content-Length: 120
Last-Modified: Thu, 19 Mar 2026 07:59:15 GMT
ETag: "69bbacd3-78"
Accept-Ranges: bytes
Cache-Control: Idco-NGFW;detail=mismatch
Connection: keep-alive

(ansible) [user@adm-hq ansible]$ curl -I http://srv3-cod.au.team
HTTP/1.1 200 OK
Server: Angie/1.10.2
Date: Thu, 19 Mar 2026 08:00:17 GMT
Content-Type: text/html
Content-Length: 120
Last-Modified: Thu, 19 Mar 2026 07:59:15 GMT
ETag: "69bbacd3-78"
Accept-Ranges: bytes
Cache-Control: Idco-NGFW;detail=mismatch
Connection: keep-alive

(ansible) [user@adm-hq ansible]$
```

```
(ansible) [user@adm-hq ansible]$ curl http://srv1-cod.au.team
<html>
  <head>
    <title>AU_Team</title>
  </head>
  <body>
    <h1>srv1-cod by Angie!</h1>
  </body>
</html>
(ansible) [user@adm-hq ansible]$ curl http://srv2-cod.au.team
<html>
  <head>
    <title>AU_Team</title>
  </head>
  <body>
    <h1>srv2-cod by Angie!</h1>
  </body>
</html>
(ansible) [user@adm-hq ansible]$ curl http://srv3-cod.au.team
<html>
  <head>
    <title>AU_Team</title>
  </head>
  <body>
    <h1>srv3-cod by Angie!</h1>
  </body>
</html>
(ansible) [user@adm-hq ansible]$
```

- Запускаем данный playbook повторно:
 - все задачи должны быть **OK**

```
(ansible) [user@adm-hq ansible]$ ansible-playbook -i inventories/production/hosts playbook2_web.yml

PLAY [Install Installing the Angie Web Server] *****

TASK [Gathering Facts] *****
ok: [srv2-cod]
ok: [srv3-cod]
ok: [srv1-cod]

TASK [Install package 'angie'] *****
ok: [srv1-cod]
ok: [srv3-cod]
ok: [srv2-cod]

TASK [Copy the 'index.html' file] *****
ok: [srv3-cod]
ok: [srv2-cod]
ok: [srv1-cod]

TASK [Started and enabled angie] *****
ok: [srv3-cod]
ok: [srv2-cod]
ok: [srv1-cod]

PLAY RECAP *****
srv1-cod      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
srv2-cod      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
srv3-cod      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

(ansible) [user@adm-hq ansible]$
```

- Находясь с требуемой директории с активированным виртуальным окружением
 - создаём файл **playbook3_haproxy.yml**
 - и описываем весь необходимый функционал


```
cat <<EOF > playbook3_haproxy.yml
---
- name: Install and settings haproxy for HA1-COD and HA2-COD
  hosts: proxy
  become: true

  tasks:
    - name: Install package 'haproxy'
      community.general.apt_rpm:
        name: "haproxy"
        state: present
        update_cache: true

    - name: Copy the 'haproxy.cfg' file
      ansible.builtin.template:
        src: templates/haproxy.cfg.j2
        dest: /etc/haproxy/haproxy.cfg
        owner: root
        group: root
        mode: '0644'

    - name: Started and enabled haproxy
      ansible.builtin.systemd:
        name: haproxy
        state: started
        enabled: true
EOF
```

- создаём необходимый шаблон:

```

cat <<EOF > templates/haproxy.cfg.j2
global
    chroot      /var/lib/haproxy
    pidfile     /var/run/haproxy.pid
    maxconn     4000
    daemon

defaults
    log         global
    mode        http
    retries     2
    timeout client 30m
    timeout connect 4s
    timeout server 30m
    timeout check 5s

frontend main
    bind {{ haproxy_frontend_bind_address }}:{{ haproxy_frontend_bind_port }}
    default_backend      app

backend app
    balance      roundrobin
    option httpchk GET /
    http-request set-header X-Forwarded-For %[src]
    http-request set-header X-Forwarded-Proto http
{% for record in haproxy_backend_add_hosts %}
    server {{ record.name }} {{ record.address }}:80 check
{% endfor %}

listen stats
    bind *:9000
    mode http
    stats enable
    stats hide-version
    stats realm Haproxy\ Statistics
    stats uri /haproxy_stats
EOF

```

- Добавляем необходимые переменные с их значениями:

```

cat <<EOF >> inventories/production/group_vars/proxy.yml

haproxy_frontend_bind_address: "0.0.0.0"
haproxy_frontend_bind_port: "80"
haproxy_backend_add_hosts:
  - name: "srv1-cod"
    address: "172.16.1.1"
  - name: "srv2-cod"
    address: "172.16.1.2"
  - name: "srv3-cod"
    address: "172.16.1.3"
EOF

```

- Запускаем данный playbook:

```

ansible-playbook -i inventories/production/hosts playbook3_haproxy.yml

```

- Результат:

```

(ansible) [user@adm-hq ansible]$ ansible-playbook -i inventories/production/hosts playbook3_haproxy.yml

PLAY [Install and settings haproxy for HA1-COD and HA2-COD] *****

TASK [Gathering Facts] *****
ok: [ha2-cod]
ok: [ha1-cod]

TASK [Install package 'haproxy'] *****
changed: [ha1-cod]
changed: [ha2-cod]

TASK [Copy the 'haproxy.cfg' file] *****
changed: [ha2-cod]
changed: [ha1-cod]

TASK [Started and enabled haproxy] *****
changed: [ha2-cod]
changed: [ha1-cod]

PLAY RECAP *****
ha1-cod      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ha2-cod      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

(ansible) [user@adm-hq ansible]$

```

- Проверить:

```

(ansible) [user@adm-hq ansible]$ curl http://172.16.1.253
<html>
  <head>
    <title>AU_Team</title>
  </head>
  <body>
    <h1>srv1-cod by Angie!</h1>
  </body>
</html>
(ansible) [user@adm-hq ansible]$ curl http://172.16.1.253
<html>
  <head>
    <title>AU_Team</title>
  </head>
  <body>
    <h1>srv2-cod by Angie!</h1>
  </body>
</html>
(ansible) [user@adm-hq ansible]$ curl http://172.16.1.253
<html>
  <head>
    <title>AU_Team</title>
  </head>
  <body>
    <h1>srv3-cod by Angie!</h1>
  </body>
</html>
(ansible) [user@adm-hq ansible]$

```

HAProxy

Statistics Report for pid 3194

> General process information

pid = 3194 (process #1, nbproc = 1, nbthread = 1)
 uptime = 0d 0h02m14s; warnings = 0
 system limits: memmax = unlimited; ulimit-n = 8051
 maxsock = 8051; maxconn = 4000; reached = 0; maxpipes = 0
 current conns = 1; current pipes = 0/0; conn rate = 1/sec; bit rate = 0.271 kbps
 Running tasks: 0/22 (0 niced); idle = 100 %

active UP backup UP
 active UP, going down backup UP, going down
 active DOWN, going up backup DOWN, going up
 active or backup DOWN not checked
 active or backup DOWN for maintenance (MAINT)
 active or backup SOFT STOPPED for maintenance
 Note: "NOLB"/"DRAIN" = UP with load-balancing disabled.

main																			
	Queue			Session rate			Sessions						Bytes		Denied		Errors		
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp
Frontend				0	1	-	0	2	4 000	3			318	960	0	0	0		

app	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings	
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	R
srv1-cod	0	0	-	0	1	0	0	1	-	1	1	55s	106	320		0		0	0	0	0
srv2-cod	0	0	-	0	1	0	0	1	-	1	1	52s	106	320		0		0	0	0	0
srv3-cod	0	0	-	0	1	0	0	1	-	1	1	50s	106	320		0		0	0	0	0
Backend	0	0		0	1	0	0	1	400	3	3	50s	318	960	0	0		0	0	0	0

stats																				
	Queue			Session rate			Sessions						Bytes		Denied		Errors			
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	
Frontend				1	1	-	1	1	1	4 000	1		0	0	0	0	0		0	
Backend	0	0		0	0		0	0		400	0	0	0s	0	0	0	0	0	0	

- Запускаем данный playbook повторно:
 - все задачи должны быть OK

```
(ansible) [user@adm-hq ansible]$ ansible-playbook -i inventories/production/hosts playbook3_haproxy.yml

PLAY [Install and settings haproxy for HA1-COD and HA2-COD] *****

TASK [Gathering Facts] *****
ok: [ha2-cod]
ok: [ha1-cod]

TASK [Install package 'haproxy'] *****
ok: [ha2-cod]
ok: [ha1-cod]

TASK [Copy the 'haproxy.cfg' file] *****
ok: [ha2-cod]
ok: [ha1-cod]

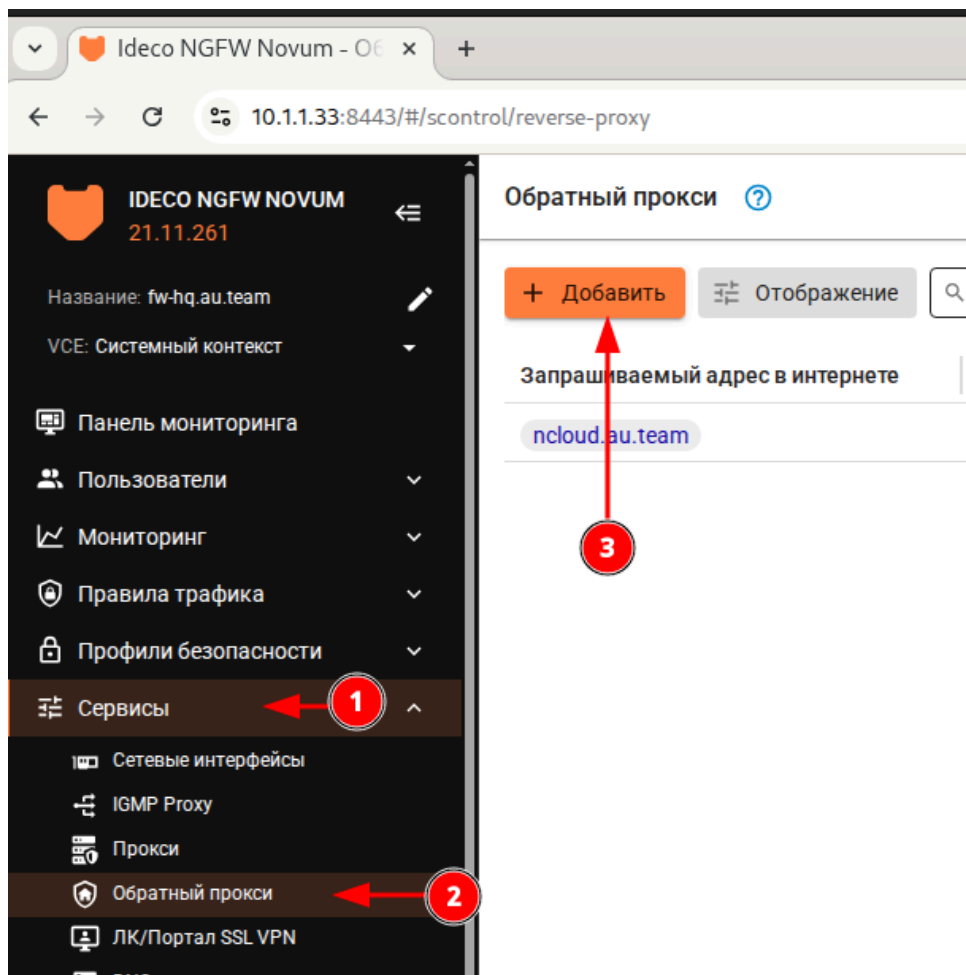
TASK [Started and enabled haproxy] *****
ok: [ha1-cod]
ok: [ha2-cod]

PLAY RECAP *****
ha1-cod      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ha2-cod      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

(ansible) [user@adm-hq ansible]$
```

ADM-HQ:

- Добавляем обратный прокси для безопасной публикации в рамках локальной сети:



- заполняем форму Добавление правила публикации

Основные настройки

Запрашиваемый адрес в интернете

www.au.team

Формат: IP-адрес, доменное имя или URL (без протокола)

+ Добавить адрес

☐ Внутренний сервис Ideco NGFW Novum

Адреса web-серверов для балансировки запросов между ними

Протокол HTTP

Используется для всех адресов

Конечный адрес 172.16.1.253

Адрес, на который будут перенаправлены запросы
Можно указать: IP:порт, домен:порт, IP или домен

Добавить адрес web-сервера

Путь

Поле необязательное. Используется для всех адресов

Настройки безопасности

Профиль WAF

☐ Включить защиту от DoS-атак

Лимит запросов с одного IP в секунду 200

Лимит подключений с одного IP 1000

Максимальный размер запроса, МБ 50

Дополнительные настройки

- ☒ Перенаправлять HTTP запросы на HTTPS
- Должно получиться следующее:
 - требуемое имя и автоматический редирект запросов с http на https

+ Добавить Отображение Поиск

Запрашиваемый адрес в интернете	Внутре...	Конечный адрес	Профи...	Тип пу...	Дополнительные настройки	Защит...	Комме...	Управление
www.au.team	—	http://172.16.1.253	—	Станд...	Перенаправлять HTTP запросы на HTTPS	—		🔌 ✎ 🗑
ncloud.au.team	—	http://srv-br.au.team	—	Станд...	Перенаправлять HTTP запросы на HTTPS	—		🔌 ✎ 🗑

- а также и корректный сертификат:

ДЕЙСТВУЮЩИЕ СЕРТИФИКАТЫ

ЗАГРУЖЕННЫЕ СЕРТИФИКАТЫ

СИСТЕМНЫЕ СЕРТИФИКАТЫ

Фильтры Отображение Поиск

Статус	Домен	Тип	Издатель
●	Ideco NGFW (Корневой)	Автоматически сгенерированный	Ideco NGFW
●	10.0.1.1	Автоматически сгенерированный	Ideco NGFW
●	10.0.2.1	Автоматически сгенерированный	Ideco NGFW
●	10.1.1.1	Автоматически сгенерированный	Ideco NGFW
●	10.1.1.33	Автоматически сгенерированный	Ideco NGFW
●	10.1.2.1	Автоматически сгенерированный	Ideco NGFW
●	63.27.18.18	Автоматически сгенерированный	Ideco NGFW
●	agent.local	Автоматически сгенерированный	Ideco NGFW
●	ncloud.au.team	Автоматически сгенерированный	Ideco NGFW
●	web-interface.local	Автоматически сгенерированный	Ideco NGFW
●	www.au.team	Автоматически сгенерированный	Ideco NGFW

- На DNS во FreeIPA добавляем CNAME запись следующего вида:

Добавить запись ресурса DNS

Имя записи *

www

Тип записи

CNAME

Hostname ? *

fw-hq.au.team.

* Обязательное поле

Добавить

Добавить и добавить ещё

Добавить и изменить

Отменить

- Результат:

www	CNAME	fw-hq.au.team.
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- Проверяем доступ по <https://www.au.team>:

srv1-cod

au.team x

 Подключение защищено >

 Файлы cookie и данные сайтов >

 Настройки сайтов 